

Homework - 8

Solⁿ ①.

given.

① Elastic constitutive law:

$$\dot{\sigma}_{ij} = 2\mu(\dot{\epsilon}_{ij} - \dot{\epsilon}_{ij}^p) + \lambda \dot{\epsilon}_{kk} \delta_{ij}$$

② yield criteria:

$$f(\underline{\sigma}, \underline{\epsilon}) = \sigma_{eq} - \sigma_y(\underline{\epsilon}) \leq 0$$

$$\text{where, } \sigma_{eq} = \sqrt{\frac{2}{3} \beta_{ij} \beta_{ij}}$$

$$\beta_{ij} = \tau_{ij} - \alpha_{ij}$$

③ flow rule:

$$\dot{\epsilon}_{ij}^p = \dot{\lambda} \frac{\partial f}{\partial \sigma_{ij}}$$

Now

$$\frac{\partial f}{\partial \sigma_{ij}} = \frac{\partial}{\partial \sigma_{ij}} \left(\sqrt{\frac{2}{3} \beta_{ke} \beta_{ke}} \right) = \frac{1}{2} \cdot \frac{1}{\sqrt{\frac{2}{3} \beta_{mn} \beta_{mn}}} \cdot \frac{2}{2} \frac{\partial (\beta_{ke} \beta_{ke})}{\partial \sigma_{ij}}$$

$$= \frac{1}{2} \cdot \frac{1}{\sigma_{eq}} \cdot \frac{2}{2} \cdot \beta_{ke} \frac{\partial \beta_{ke}}{\partial \sigma_{ij}}$$

$$= \frac{3}{2} \frac{\beta_{ke}}{\sigma_{eq}} \frac{\partial \beta_{ke}}{\partial \sigma_{ij}} \quad \text{--- ①}$$

Now

$$\beta_{ke} = \sigma_{ke} - \sigma_m \delta_{ke} - \alpha_{ke}$$

$$\frac{\partial \beta_{ke}}{\partial \sigma_{ij}} = \delta_{ik} \delta_{je} - \frac{1}{3} \delta_{ij} \delta_{ke} - \frac{2}{3} K \left(\dot{\lambda} \frac{\partial}{\partial \sigma_{ij}} \cdot \frac{\partial f}{\partial \sigma_{ke}} \right) \xrightarrow{\text{ignore high order term.}}$$

Putting ② in ①

$$= \frac{\sigma}{2} \frac{\beta_{k\ell}}{\sigma_{eq}} \cdot (\delta_{ik}\delta_{j\ell} - \frac{1}{2}\delta_{k\ell}\delta_{ij})$$

$$= \frac{\sigma}{2} \left(\frac{\beta_{ij}}{\sigma_{eq}} - \frac{1}{3} \frac{\beta_{kik\ell}\delta_{ij}}{\sigma_{eq}} \right) \quad \text{--- (3)}$$

Now

$$\beta_{kik\ell} = \tau_{kik\ell} - \alpha_{kik\ell} = 0$$

$$\therefore \text{tr}(\underline{\underline{\tau}}) = 0$$

$$\text{tr}(\underline{\underline{\dot{\epsilon}}^p}) = 0$$

$\therefore$ putting ③ in flow rule.

$$\boxed{\dot{\epsilon}_{ij}^p = \frac{\sigma}{2} \lambda \frac{\beta_{ij}}{\sigma_{eq}}} \quad \text{--- (4)}$$

Now

④ Hardening.

↳ kinematic hardening

$$\dot{\alpha} = \frac{2}{3} K \dot{\epsilon}_{ij}^p = K \lambda \frac{\beta_{ij}}{\sigma_{eq}} \quad \text{--- (5)}$$

↳ isotropic hardening.

$$\sigma_y(s), \quad \dot{s} = \sqrt{\frac{2}{3} \dot{\epsilon}_{ij}^p \dot{\epsilon}_{ij}^p}$$

$$\dot{s} = \frac{\sigma}{2} \lambda \frac{\sqrt{\beta_{ij}\beta_{ij}}}{\sigma_{eq}} \cdot \frac{2}{3} \cdot \frac{\sigma}{2} \cdot \frac{2}{3}$$

$$\boxed{\dot{s} = \lambda} \quad \text{--- (6)}$$

Now
⑤

Lahti-Tucker condition.

$$\dot{\lambda} \geq 0 \quad \dot{f} \leq 0 \quad \dot{\lambda} \dot{f} = 0$$

⑥ consistency condition:

$$\text{if } \dot{\lambda} > 0, \quad \dot{f} = 0$$

$$f = \sqrt{\frac{g}{2} \beta_{ij} \beta_{ij}} = \sigma_y(1)$$

$$\dot{f} = \frac{g}{2} \frac{\beta_{ij}}{\sigma_y} \dot{\beta}_{ij} - \frac{d\sigma_y}{ds} \dot{s} = 0 \quad \text{--- (6.1)}$$

$$\text{where } h(s) = \frac{d\sigma_y}{ds}$$

$$\Rightarrow \frac{g}{2} \frac{\beta_{ij}}{\sigma_y} \dot{\beta}_{ij} - h \dot{s} = 0$$

Now

$$\dot{\beta}_{ij} = \dot{\epsilon}_{ij} - \dot{\alpha}_{ij} = \dot{\sigma}_{ij} - \frac{1}{3} \dot{\sigma}_{kk} \delta_{ij} - \dot{\alpha}_{ij} \quad \text{--- (7)}$$

Now

$$\dot{\sigma}_{ij} = 2\mu(\dot{\epsilon}_{ij} - \dot{\epsilon}_{ij}^p) + \bar{\lambda} \dot{\epsilon}_{kk} \delta_{ij}$$

$$\dot{\sigma}_{kk} = 2\mu(\dot{\epsilon}_{kk}) + \bar{\lambda} \dot{\epsilon}_{kk} \cdot 3 = 2\mu \dot{\epsilon}_{kk} + 3\bar{\lambda} \dot{\epsilon}_{kk}$$

$$= (2\mu + 3\bar{\lambda}) \dot{\epsilon}_{kk}$$

$$\dot{\alpha}_{ij} = \frac{2\mu}{3} \dot{\epsilon}_{ij}^p$$

} --- (8)

putting (8) in (7)

$$\dot{\beta}_{ij} = 2\mu(\dot{\epsilon}_{ij} - \dot{\epsilon}_{ij}^p) + \cancel{\lambda \dot{\epsilon}_{klc} \delta_{ij}} - \frac{1}{3}(2\mu + 8\lambda) \cancel{\dot{\epsilon}_{klc} \delta_{ij}} - \frac{2}{3} \kappa \dot{\epsilon}_{ij}^p$$

$$\dot{\beta}_{ij} = 2\mu(\dot{\epsilon}_{ij} - \dot{\epsilon}_{ij}^p) - \frac{2}{3} \mu \dot{\epsilon}_{klc} \delta_{ij} - \frac{2}{3} \kappa \dot{\epsilon}_{ij}^p \quad (9)$$

~~putting (9) in (6.1)~~

putting (9) in (9)

$$\dot{\beta}_{ij} = 2\mu \dot{\epsilon}_{ij} - \frac{3\mu \dot{\lambda}}{\sigma_{eq}} \beta_{ij} - \frac{2}{3} \mu \dot{\epsilon}_{klc} \delta_{ij} - \frac{\dot{\lambda} \kappa}{\sigma_{eq}} \beta_{ij} \quad (10)$$

Now, putting (10) in (6.1)

$$\frac{3}{2} \frac{\beta_{ij}}{\sigma_{eq}} \left[2\mu \dot{\epsilon}_{ij} - \frac{3\mu \dot{\lambda}}{\sigma_{eq}} \beta_{ij} - \frac{2}{3} \mu \dot{\epsilon}_{klc} \delta_{ij} - \frac{\dot{\lambda} \kappa}{\sigma_{eq}} \beta_{ij} \right] - h \dot{\lambda} = 0$$

$$\frac{1}{\sigma_{eq}} \left[3\mu \dot{\epsilon}_{ij} \frac{\beta_{ij}}{\sigma_{eq}} - \frac{3\mu \dot{\lambda} \sigma_{eq}^2}{\sigma_{eq}} - \frac{2}{3} \mu \dot{\epsilon}_{klc} \beta_{ij} - \frac{\dot{\lambda} \kappa \beta_{ij} \sigma_{eq}^2}{\sigma_{eq}} \right] - h \dot{\lambda} = 0$$

$$\frac{3\mu \dot{\epsilon}_{ij} \beta_{ij}}{\sigma_{eq}} - 3\mu \dot{\lambda} - \dot{\lambda} \kappa - h \dot{\lambda} = 0$$

$$\dot{\lambda} = \frac{3\mu \beta_{ij} \dot{\epsilon}_{ij}}{\sigma_{eq} [3\mu + \kappa + h]} \quad (11)$$

Now, $\dot{\lambda} > 0$

$$\dot{\sigma}_{ij} = C_{ijkl} \dot{\epsilon}_{kl}$$

$$\dot{\sigma}_{ij} = 2\mu (\dot{\epsilon}_{ij} - \dot{\epsilon}_{ij}^p) + \bar{\lambda} \dot{\epsilon}_{kk} \delta_{ij}$$

$$\dot{\sigma}_{ij} = 2\mu \left(\dot{\epsilon}_{ij} - \frac{g}{2} \frac{\dot{\lambda} \beta_{ij}}{\sigma_{eq}} \right) + \bar{\lambda} \dot{\epsilon}_{kk} \delta_{ij}$$

$$\dot{\sigma}_{ij} = \underbrace{2\mu \dot{\epsilon}_{ij} + \bar{\lambda} \dot{\epsilon}_{kk} \delta_{ij}}_{C_{ijkl} \dot{\epsilon}_{kl}} - \frac{2\mu \dot{\lambda} \beta_{ij}}{\sigma_{eq}}$$

$$\dot{\sigma}_{ij} = C_{ijkl} \dot{\epsilon}_{kl} - \frac{2\mu \dot{\lambda} \beta_{ij}}{\sigma_{eq}} \quad (12)$$

Now put (11) in (12)

$$\dot{\sigma}_{ij} = C_{ijkl} \dot{\epsilon}_{kl} - \frac{g}{\sigma_{eq}} \left[\frac{3\mu^2 \beta_{kl} \dot{\epsilon}_{kl}}{\sigma_{eq} (3\mu + k + h)} \right] \beta_{ij}$$

$$\dot{\sigma}_{ij} = \left[C_{ijkl} - \frac{9\mu^2 \beta_{ij} \beta_{kl}}{\sigma_{eq}^2 (3\mu + k + h)} \right] \dot{\epsilon}_{kl}$$

from $f = 0$, $\sigma_{eq} - \sigma_y = 0$, $\sigma_{eq} = \sigma_y$

$$\therefore \boxed{C_{ijkl}^t = C_{ijkl} - \frac{9\mu^2 \beta_{ij} \beta_{kl}}{\sigma_y^2 (3\mu + k + h)}}$$

Soln ② governing eqn

① Elastic law:

$$\delta \sigma_{ij} = 2\mu(\delta \epsilon_{ij} - \delta \epsilon_{ij}^p) + \bar{\lambda} \delta \epsilon_{kk} \delta_{ij}$$

② yield criteria:

$$f(\tau_{ij} + \delta \tau_{ij}, s + \delta s) \leq 0$$

$$\text{i.e. } \left[\frac{s}{2} (\tau_{ij} + \delta \tau_{ij}) (\tau_{ij} + \delta \tau_{ij}) \right]^{1/2} - \sigma_y (s + \delta s) \leq 0$$

$$\text{where } \sigma_y(s) = \sigma_0 (1 + s)^\eta$$

$$\delta s = \sqrt{\frac{2}{3} \delta \epsilon_{ij}^p \delta \epsilon_{ij}^p} = \delta \lambda$$

③ Flow rule:

$$\delta \epsilon_{ij}^p = \frac{s}{2} \frac{\delta \lambda}{\sigma_y(s + \delta \lambda)} (\tau_{ij} + \delta \tau_{ij})$$

④ Drucker P-T condition:

$$\text{if } \delta \lambda > 0, f(\tau_{ij} + \delta \tau_{ij}, s + \delta s) = 0$$

Now

Elastic prediction:-

$$\sigma^t = 2\mu \delta \epsilon_{ij} + \bar{\lambda} \delta \epsilon_{kk} \delta_{ij}, \text{ assuming } \delta \lambda = 0$$

$$\delta \epsilon_{ij}^p = 0, \delta s = 0$$

$$\text{if } f^t = f(\tau_{ij} + \delta \tau_{ij}^t, s) \leq 0$$

$$\text{i.e. } \delta \tau_{ij}^t = 2\mu \delta \epsilon_{ij} - \frac{2}{3} \mu \delta \epsilon_{kk} \delta_{ij}$$

$$\text{if } f^t \leq 0 \quad \delta \lambda = 0 \quad \delta \sigma_{ij} = \delta \sigma^t, \delta \epsilon_{ij}^p = 0$$

Now~~Equation~~if $\delta\lambda > 0$ find $\delta\lambda$.

$$\phi(\tau_{ij} + \delta\tau_{ij}, s + \delta s) = 0$$

$$\delta\tau_{ij} = 2\mu (\epsilon_{ij} - \delta\epsilon_{ij}^p) - \frac{\sigma}{2} \mu \epsilon_{kk} \delta\epsilon_{ij}$$

$$\delta\tau_{ij} = \delta\tau_{ij}^t - 2\mu \delta\epsilon_{ij}^p \quad \text{--- (1)}$$

$$\text{and } \delta\epsilon_{ij}^p = \frac{\sigma}{2} \frac{\delta\lambda}{\sigma_y(s + \delta\lambda)} (\tau_{ij} + \delta\tau_{ij}) \quad \text{--- (2)}$$

Now

$$\tau_{ij} + \delta\tau_{ij} = \tau_{ij} + \delta\tau_{ij}^t - 2\mu \delta\epsilon_{ij}^p \quad \text{--- (3)}$$

From (2) & (3)

$$\tau_{ij} + \delta\tau_{ij} = \tau_{ij} + \delta\tau_{ij}^t - 2\mu \cdot \frac{\sigma}{2} \frac{\delta\lambda}{\sigma_y(s + \delta\lambda)} (\tau_{ij} + \delta\tau_{ij})$$

$$(\tau_{ij} + \delta\tau_{ij}) \left[1 + \frac{3\mu\delta\lambda}{\sigma_y(s + \delta\lambda)} \right] = \tau_{ij} + \delta\tau_{ij}^t$$

$$\tau_{ij} + \delta\tau_{ij} = \frac{\tau_{ij} + \delta\tau_{ij}^t}{\left[1 + \frac{3\mu\delta\lambda}{\sigma_y(s + \delta\lambda)} \right]} \quad \text{--- (4')}$$

Now from (4)

$$\phi(\tau_{ij} + \delta\tau_{ij}, s + \delta s) = 0 \quad \text{for } \delta\lambda > 0$$

$$\frac{1}{\left[1 + \frac{3\mu\delta\lambda}{\sigma_y(s + \delta\lambda)} \right]} \left(\frac{3}{2} (\tau_{ij} + \delta\tau_{ij}^t) (\tau_{ij} + \delta\tau_{ij}^t) \right)^{1/2} - \sigma_y(s + \delta\lambda) = 0$$

(3)

$$\left[\frac{2}{2} (\tau_{ij} + \delta \tau_{ij}^t) (\tau_{ij} + \delta \tau_{ij}^t) \right]^{1/2} - \sigma_y (s + \delta \lambda) - 3\mu \delta \lambda = 0$$

$$* - \sigma_y(s) + \sigma_y(s)$$

$$\left[\tau^t - \sigma_y(s + \delta \lambda) + \sigma_y(s) - 3\mu \delta \lambda = 0 \right] \quad (5')$$

solving eqn (5) using numerical method

to find $\delta \lambda$

update state variables.

$$(1) \quad s_{n+1} = s_n + \delta \lambda$$

$$(2) \quad \delta \epsilon_{ij}^p = \frac{2}{3} \frac{\delta \lambda}{\sigma_y(s + \delta \lambda)} (\tau_{ij} + \delta \tau_{ij}^t)$$

$$(3) \quad \tau_{ij} + \delta \tau_{ij}^t = \frac{\tau_{ij} + \delta \tau_{ij}^t}{\left[1 + \frac{3\mu \delta \lambda}{\sigma_y(s + \delta \lambda)} \right]}$$

$$(4) \quad \delta \tau_{ij} = 2\mu (\delta \epsilon_{ij} - \delta \epsilon_{ij}^p) - \frac{2}{3} \mu \delta \epsilon_{kk} \delta \tau_{ij}$$

$$(5) \quad \delta \sigma_{ij} = c_{ijkl} (\delta \epsilon_{kl} - \delta \epsilon_{kl}^p)$$

$$(6) \quad \delta \sigma_{ij} = 2\mu (\delta \epsilon_{ij} - \delta \epsilon_{ij}^p) + \bar{\lambda} \delta \epsilon_{kk} \delta \tau_{ij}$$

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```
In [46]: import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import newton

# Material and simulation constants
nu = 0.3
E = 100000
sigma_0 = E / 500

# Lamé constants
mu = E / (2 * (1 + nu)) # Shear modulus
lambda_prime = E*nu / ((1+nu)* (1- 2 * nu))

# Plasticity parameters
n_values = [0, 0.1]
load_paths = [(-1, 0), (-0.5, -0.5)]

# Deformation range
eps_values = np.linspace(0, 0.05, 1000)
```

```
In [47]: # Deviatoric function
def deviatoric(tensor):
    trace = np.trace(tensor)
    identity = np.identity(3)
    return tensor - (trace / 3) * identity

# Yield stress in term of plastic arc length and n
def yield_stress(s, n):
    return sigma_0 * (1 + 500 * s) ** n
```

```
In [48]: def radial_return(sigma_trail, s, n):
    sigma_trail_dev = deviatoric(sigma_trail)
    sigma_eq_trail = np.sqrt((3/2) * np.sum(sigma_trail_dev**2))
    yield_fun_trail = sigma_eq_trail - yield_stress(s,n)

    if yield_fun_trail <= 0.0:
        return np.zeros((3, 3)), 0.0 # Elastic
    else:
        # Plastic correction
        d_lambda_0 = 0

        # Define the function
        def equation(d_lambda):
            eq = yield_fun_trail + yield_stress(s,n) - yield_stress(s+d_lambda,n)
            return eq

        # Solve using Newton-Raphson method
        soln = newton(equation, d_lambda_0)
        d_lambda = soln
        s = s + d_lambda
        C = [1+(3*mu*d_lambda/yield_stress(s,n))] # dividing this with trail devi
```

```

                                # gives actual deviatoric stress
sigma_dev= sigma_trail_dev/ C
d_eps_p= (3/2)*d_lambda*sigma_dev/ yield_stress(s,n)

return d_eps_p, s

```

```

In [49]: def run_simulation(a,b,n):
          stresses = []
          plastic_strains = []
          s = 0.0
          sigma= np.zeros((3,3))
          eps_p_total = np.zeros((3, 3))

          for i in range(1, len(eps_values)):
              d_eps_value= eps_values[i]-eps_values[i-1] # change in strain value
              d_eps = np.array([[1, 0, 0],
                                [0, a, 0],
                                [0, 0, b]]) * d_eps_value # Change in strain tensor

              d_sigma_trail= 2*mu*d_eps + lambda_prime*np.trace(d_eps) * np.eye(3)
              sigma_trail= sigma + d_sigma_trail

              d_eps_p, s= radial_return(sigma_trail, s, n)
              eps_p_total = eps_p_total+ d_eps_p

              # correcting trail stress to get actual stress
              d_sigma = 2*mu*(d_eps-d_eps_p) + lambda_prime*np.trace(d_eps) * np.eye(3)
              sigma = sigma+ d_sigma

              # stresses.append(sigma)
              # plastic_strains.append(eps_p_total)

          stresses.append([sigma[0, 0], sigma[1, 1], sigma[2, 2]])
          plastic_strains.append([eps_p_total[0, 0], eps_p_total[1, 1], eps_p_total[2, 2]])

          return np.array(stresses), np.array(plastic_strains)

```

```

In [50]: # Plotting
          for a, b in load_paths:
              for n in n_values:
                  stresses, plastic_strains = run_simulation(a, b, n)
                  plt.figure()
                  plt.plot(eps_values[1:], stresses[:, 0], label='σ11')
                  plt.plot(eps_values[1:], stresses[:, 1], label='σ22')
                  plt.plot(eps_values[1:], stresses[:, 2], label='σ33')
                  plt.title(f'Stresses: a={a}, b={b}, n={n}')
                  plt.xlabel('ε')
                  plt.ylabel('Stress')
                  plt.legend()
                  plt.grid()

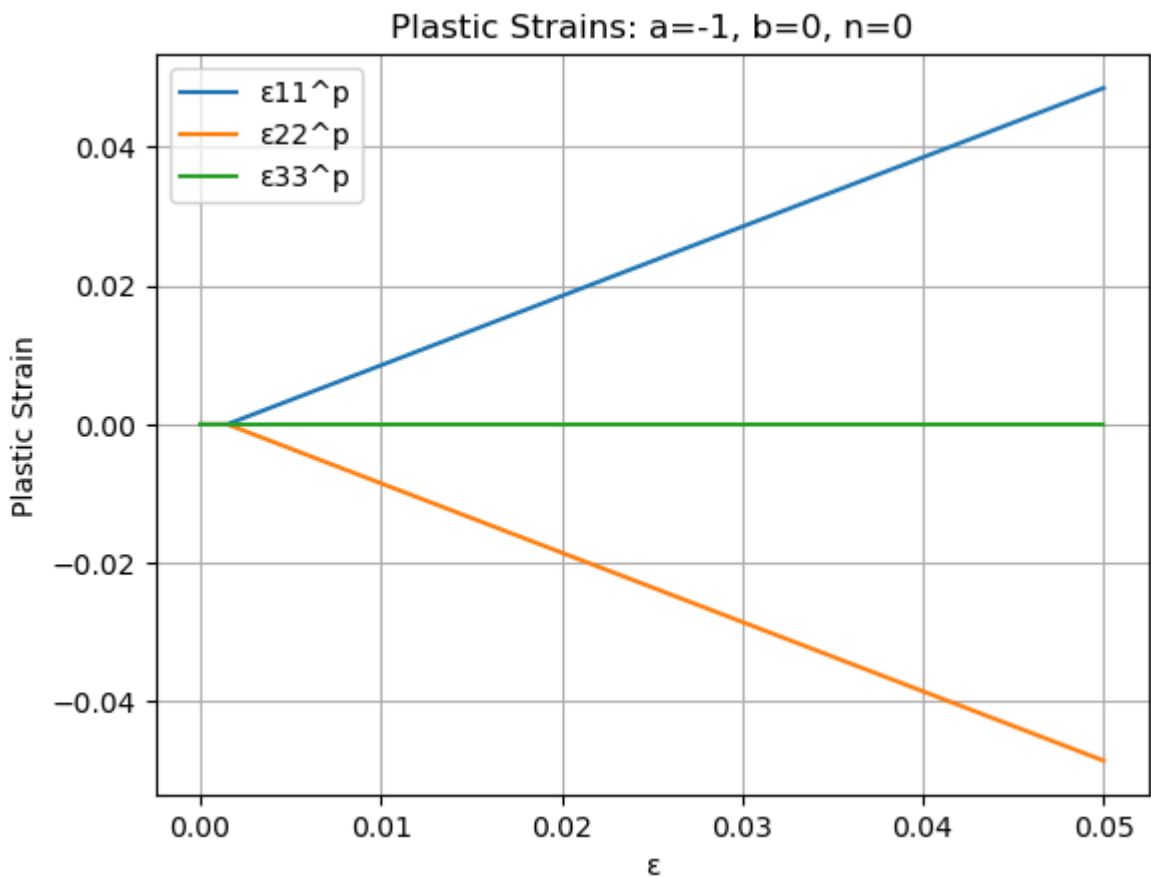
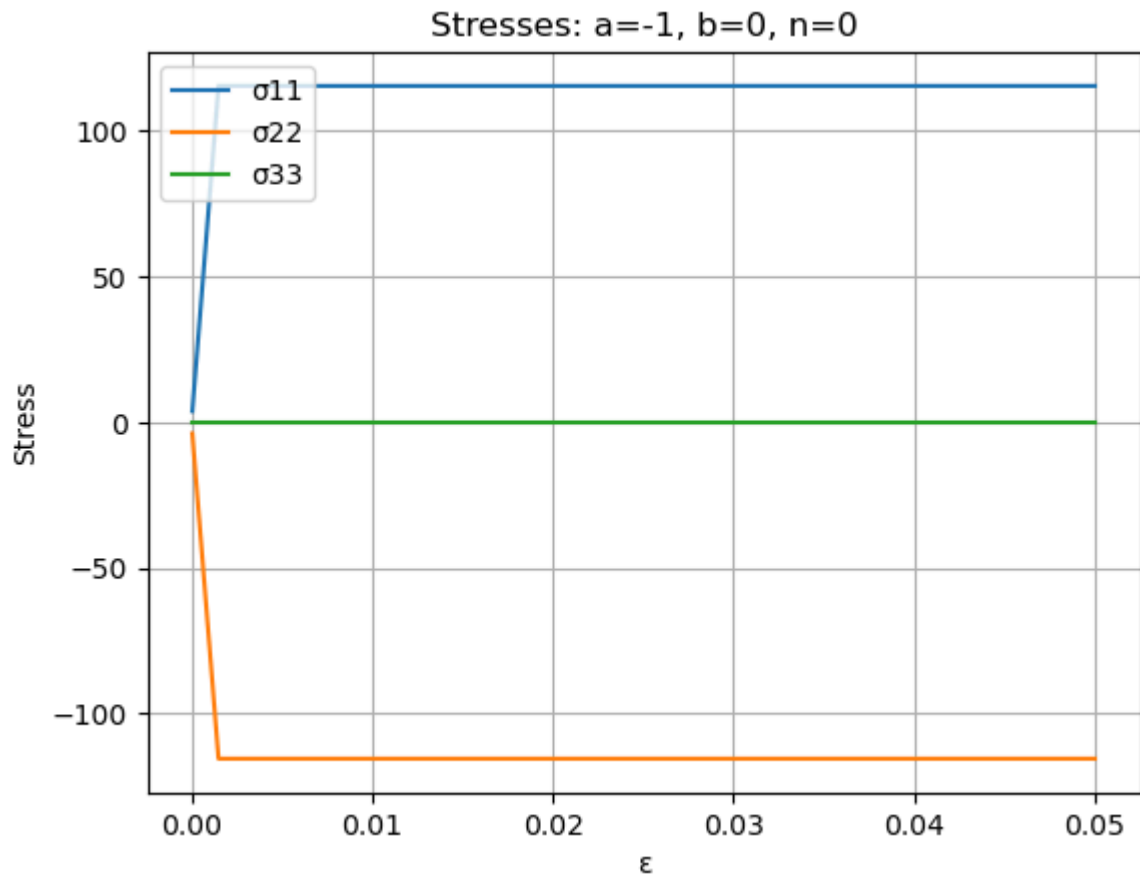
                  plt.figure()
                  plt.plot(eps_values[1:], plastic_strains[:, 0], label='ε11p')
                  plt.plot(eps_values[1:], plastic_strains[:, 1], label='ε22p')
                  plt.plot(eps_values[1:], plastic_strains[:, 2], label='ε33p')
                  plt.title(f'Plastic Strains: a={a}, b={b}, n={n}')
                  plt.xlabel('ε')
                  plt.ylabel('Plastic Strain')
                  plt.legend()

```

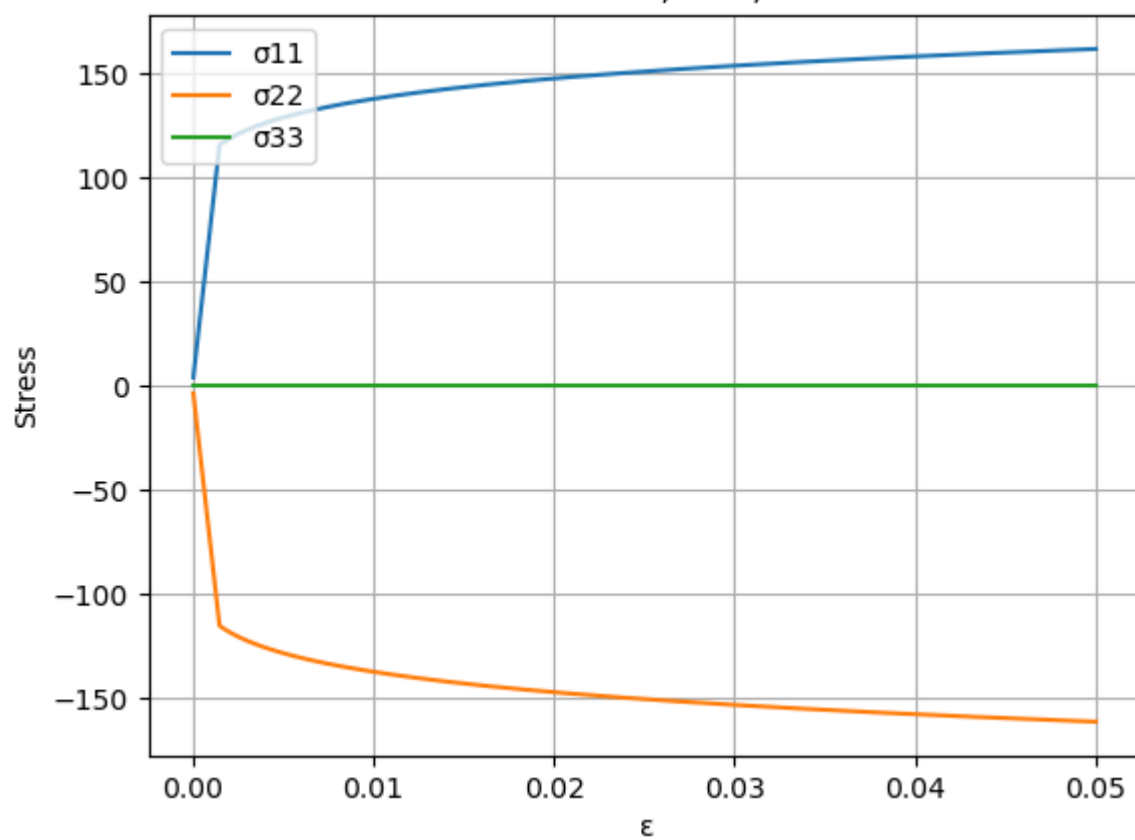


```
plt.grid()

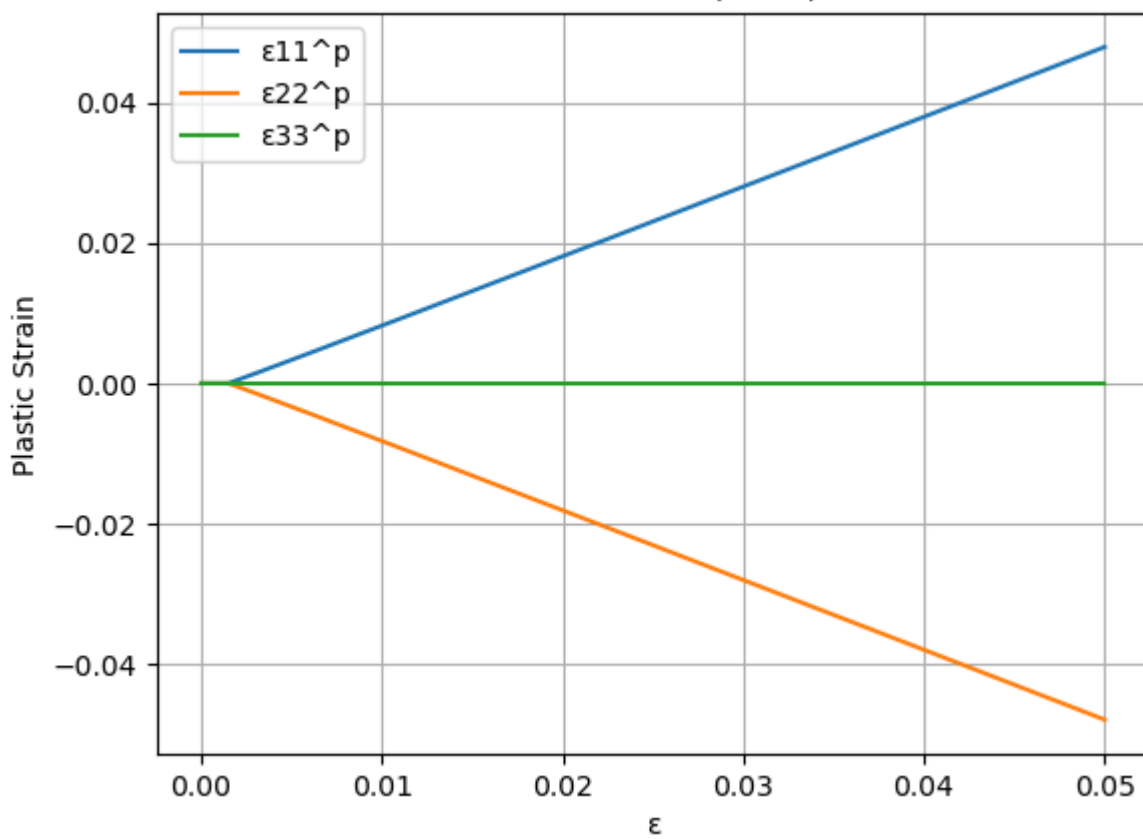
plt.show()
```



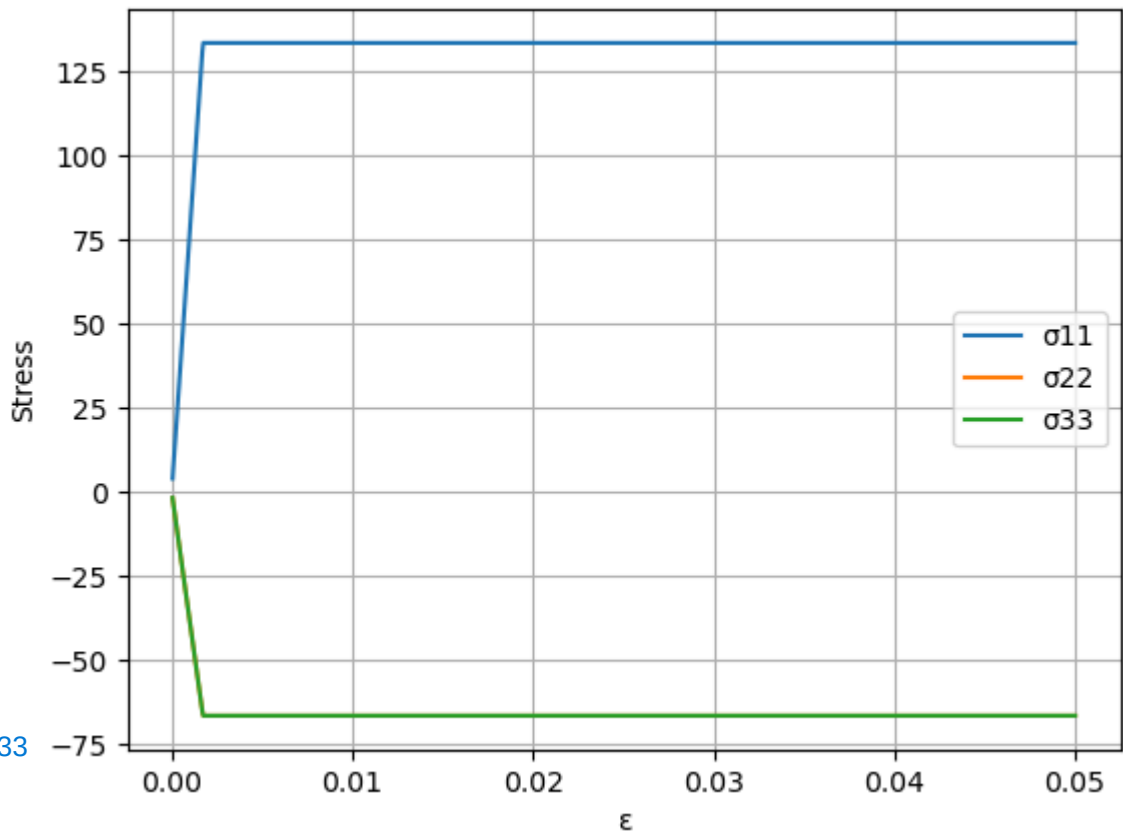
Stresses: $a=-1$, $b=0$, $n=0.1$



Plastic Strains: $a=-1$, $b=0$, $n=0.1$



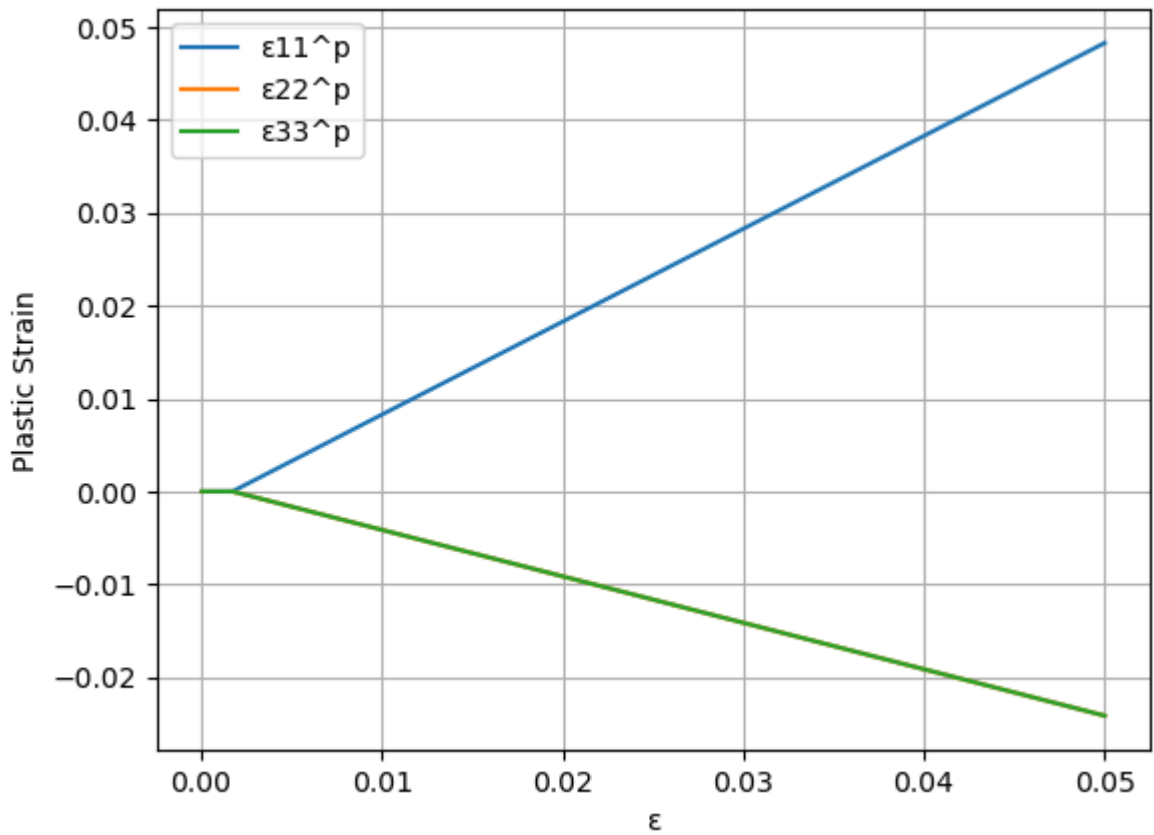
Stresses: $a=-0.5$, $b=-0.5$, $n=0$



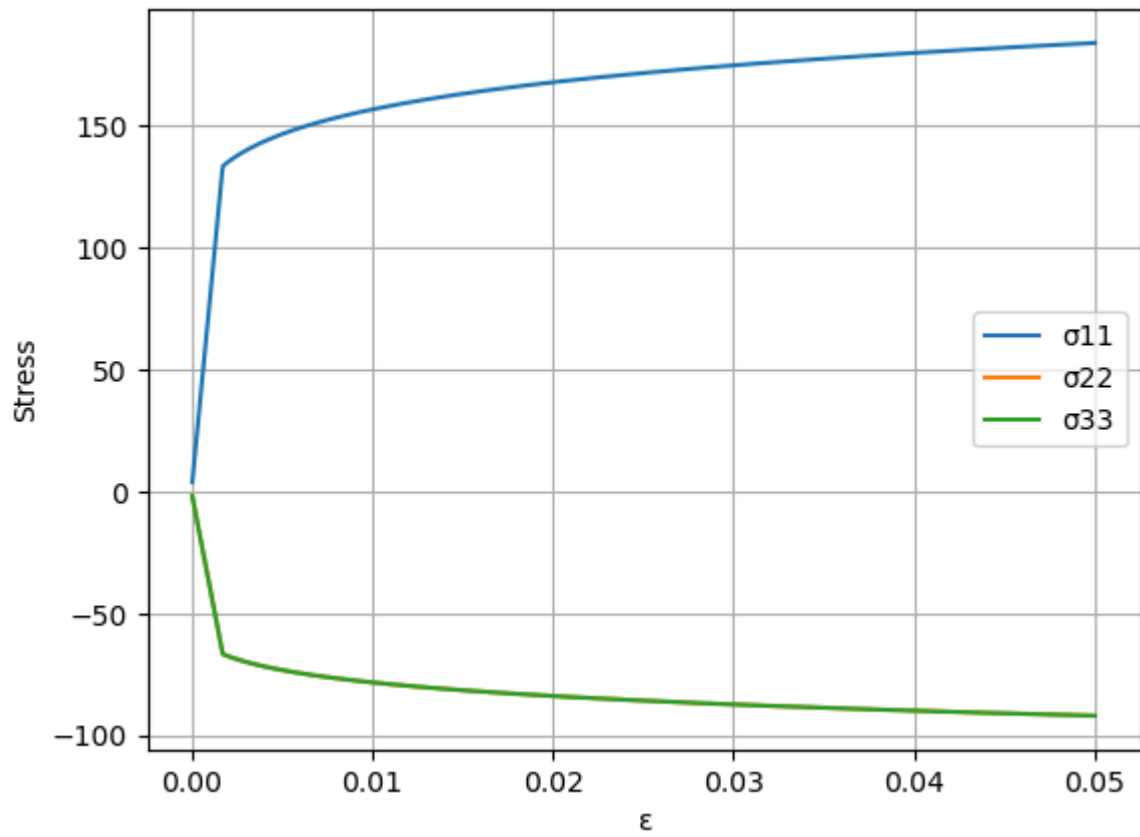
σ_{22} and σ_{33} is overlapping.

similarly,
 ϵ_{22} and ϵ_{33}
 is overlapping

Plastic Strains: $a=-0.5$, $b=-0.5$, $n=0$



Stresses: $a=-0.5$, $b=-0.5$, $n=0.1$



Plastic Strains: $a=-0.5$, $b=-0.5$, $n=0.1$

